

Motives and Role of Psychological Ownership in AR Workspaces for Remote Collaboration

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ABSTRACT

Augmented Reality (AR) systems have been proposed as solutions to reinforce remote collaboration over performing physical tasks by integrating digital information into the physical environment. However, there is a fundamental gap that arises from the difference in the degree to which remote and local workers interact with virtual and physical objects, which is attributed to differing senses of ownership over these objects. Addressing this gap, my thesis investigates how collaborative functionality affects psychological ownership in shared AR workspaces. Specifically, the study examines how the ability to modify annotations shapes Individual and Collective Psychological Ownership (IPO and CPO). Furthermore, I explore how IPO and CPO mediate the effect of collaborative functionality on the outcomes of remote collaboration. My overall goal is to extend the theoretical understanding of psychological ownership, offering insights into the dynamics of shared AR environments by focusing on the interplay between physical and virtual elements.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; *Collaborative and social computing systems and tools.*

KEYWORDS

augmented reality, remote collaboration, shared information, psychological ownership, social interaction

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1 INTRODUCTION

Augmented Reality (AR) technology seamlessly integrates digital information into the physical world, enabling individuals to interact with virtual objects within a real-world context [3]. In remote collaboration, AR technology can provide a shared virtual workspace for collaborators to work together in real-time, regardless of their location. Accordingly, a remote worker can view a local worker's

physical workspace and directly augment the local worker's view with AR technologies. As a result, both industry developers and academic researchers have begun to explore the potential of AR Head Mounted Displays (HMDs) for remote instruction [9]. For instance, Microsoft has touted the benefit of the HoloLens for remote instruction with Microsoft's Dynamics 365 Remote Assist and researchers have explored the advantages of augmenting the local worker's AR HMD view with virtual arms for a remote instructor to point to objects of interest [15, 17]. In the context of AR-supported remote collaboration, shared workspaces play a pivotal role in facilitating effective communication [14], coordination [13], and decision-making [12]. These shared workspaces create a hybrid social space, allowing multiple collaborators to interact with digital content embedded in the physical environment collectively [4]. This unique blend of physical and virtual elements gives rise to interpersonal tensions because of privacy [11] and personal space concerns [25]. One factor in understanding interpersonal tensions in shared workspaces is the concept of psychological ownership. Psychological ownership refers to the cognitive state in which individuals feel a sense of possession, control, and responsibility over objects, spaces, or ideas [22]. This sense of ownership is important for facilitating smooth communication, as it can influence how individuals interact with and relate to the shared workspace and the virtual objects within it. For example, the more emotionally attached a worker is to an object, the more they tend to protect it from others [5]. In a collaboration situation, the two forms of psychological ownership are intermixed: Individual and Collective Psychological Ownership (IPO and CPO), which is necessary to understand the reason for rising interpersonal tensions in AR shared workspaces. Individual psychological ownership refers to the sense of ownership experienced by each worker over specific objects or areas within the shared workspace, while collective psychological ownership describes the sense of ownership shared by the entire group over the workspace as a whole [20]. Striking an interplay between individual and collective ownership is essential for effective collaboration, as it allows team members to feel a sense of autonomy and personal investment while still working towards shared goals and objectives [22, 30]. Prior studies have delved into the impact of psychological ownership on collaboration within shared workspaces [7, 23, 24, 30]. To be specific, the development of psychological ownership is affected by the ability to create, change, and manipulate virtual objects [6]. For example, if a virtual object has the ability to be customized collaboratively, both participants will feel a sense of psychological ownership of the object [7]. However, those studies have not sufficiently addressed the relationship between the annotation functionality and IPO and CPO of the shared workspaces. Particularly, the activity to use annotation functionality individually or collectively within the shared workspace, affects

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shaping IPO and CPO. For example, editing one's own annotations can reinforce a sense of individual ownership, while editing others' annotations can foster a sense of collective ownership. This functionality can significantly influence collaborators' sense of ownership and control over shared resources. Addressing this gap is essential for several reasons. First, the unique characteristics of AR shared workspaces, such as the blending of virtual and physical environments, direct augmentation of users' views, and real-time interaction capabilities, may lead to different experiences of IPO and CPO compared to traditional shared workspaces. Second, the role of different annotation functionalities in shaping psychological ownership can significantly impact collaborators' sense of ownership and control over shared resources, affecting how individuals contribute to the group processes and coordinate their actions with others. Fostering a sense of ownership among remote collaborators can enhance their commitment, responsibility, and satisfaction. By identifying the factors that influence IPO and CPO in AR shared workspaces, researchers and practitioners can develop strategies to promote a sense of ownership and foster smooth communication of remote collaboration.

2 BACKGROUND AND RELATED WORK

2.1 Psychological Ownership in Shared Workspaces

Psychological ownership, a mental state where an individual perceives an object, idea, or space as "mine" [21], plays a pivotal role in shared workspaces. This sense of ownership can significantly influence individual behaviors, group dynamics, and the overall outcome of collaboration [18]. For instance, when individuals feel a sense of ownership towards their work or workspace, they tend to invest more effort, express greater creativity, and show increased commitment to their tasks, leading to enhanced job satisfaction and performance [2]. In shared workspaces, psychological ownership can shape collaboration dynamics and outcomes. Collaborators who feel a sense of ownership over the workspace or the task may exhibit higher levels of engagement, commitment, and motivation [5]. This can result in improved performance, satisfaction, and a stronger sense of belonging among team members [5]. In the context of a design team, for instance, team members who feel a sense of ownership towards the project are more likely to contribute innovative ideas and take initiative, leading to a more successful project outcome.

However, in shared AR environments, psychological ownership becomes more complex. AR technologies embed virtual objects in physical settings, creating mixed environments that encompass traits and associated meanings of both the physical and the virtual world. People may feel a sense of possession over the virtual object, leading to a unique form of psychological ownership in the virtual-physical space [7, 23]. Poretski et al. [23] outlined five key dimensions of psychological ownership in the virtual-physical space: possession, control, identity, responsibility, and territoriality. Possession, the most fundamental dimension, represents the feeling of having something [22]. Control refers to the ability to influence, manage, or manipulate the owned object or space [2, 19]. Identity signifies the extent to which the owned object or space becomes a part of the individual's self-concept [2, 6]. Responsibility involves

a sense of obligation or stewardship towards the owned object or space [2, 19, 29]. Territoriality represents protective behavior toward the target object or entity, and the tendency to defend it from others [5, 26, 28]. These dimensions of psychological ownership can significantly influence how individuals interact in shared AR workspaces, affecting their sense of ownership towards both the physical and virtual elements of the workspace [1]. It's also important to note that these dimensions are interrelated, with each one potentially influencing the others. For example, a sense of control over a virtual object might enhance a user's sense of possession and responsibility, while also influencing their territorial behavior [23].

2.2 Role of Individual and Collective Psychological Ownership in Remote Collaboration

Psychological ownership in a remote collaboration context can be classified into two main categories: individual and collective, which play different roles and can have different impacts on team dynamics and outcomes. Individual psychological ownership (IPO) is linked to the sense of possession and attachment that individual collaborators feel towards the shared task or workspace. Studies have shown that when individuals feel a strong sense of ownership towards their work, they tend to show increased motivation, engagement, and job satisfaction, which can positively influence their performance [21]. However, excessive IPO may lead to negative outcomes, such as territorial behaviors and resistance to change, particularly when it comes to shared tasks or resources. For example, a remote collaborator who feels a strong sense of ownership towards a shared document might resist changes made by others, leading to conflicts and hindering collaboration [28]. Collective psychological ownership (CPO), on the other hand, is a nuanced concept that extends beyond individual possession to a group-level phenomenon, where a collectively held understanding manifests as "ours," encompassing elements such as workspaces, projects, or ideas [20]. A strong sense of CPO can foster a shared identity, mutual trust, and cooperative behaviors among team members, which can enhance team performance and satisfaction [8, 20]. The complexity of CPO also lies in the dynamic between what is perceived as "mine" versus "ours." The ownership of certain knowledge or ideas can create a sense of superiority, leading to reluctance to share, which may engender feelings of territoriality. In collaborative environments, this can lead to conflicts between individual ownership and collective ownership, sometimes resulting in trade-offs between collaboration and the sense of psychological ownership. For instance, This tension can also affect learning environments, where assessment and evaluation of individual contributions may be challenging [16]. As a final point, both IPO and CPO play significant roles in remote collaboration, influencing individual behaviors, team dynamics, and collaboration outcomes [18]. While they can enhance engagement, performance, and satisfaction, they also present challenges that need to be carefully managed. Further research is needed to explore how these different forms of psychological ownership shape and how they can be effectively managed in remote collaboration.

2.3 Influence of AR Collaborative Functionality on Psychological Ownership

The rapid advancement of AR technology has led to an increase in interactions with virtual objects, especially in environments where the virtual layer is superimposed on the physical world. Users can assign value to these virtual objects, experience a sense of relatedness, and develop psychological ownership over these objects [23]. However, the effect of the background layer, whether physical or virtual, can significantly impact the development of relatedness and feelings of ownership, underscoring the importance of the real world in shaping perceptions [23]. In the context of AR remote collaboration, the immersive and interactive nature of AR can blur the boundaries between individual and collective ownership, creating novel challenges and opportunities for managing psychological ownership in AR remote collaboration. For example, Carrozzi et al. [7] revealed that the customization of AR holograms could foster a sense of psychological ownership over digital objects, suggesting that AR functionality allowing for personal customization can significantly influence psychological ownership. In addition, the study illustrated how the affordances of AR technology influence these mechanisms when customers switch between personal or shared devices. This finding underscores the importance of considering the specific functionality and affordances of AR technology when studying their influence on psychological ownership. As such, these studies suggest that AR functionality, such as manipulable annotations and customizable holograms, have the potential to shape psychological ownership in shared AR workspaces significantly. This shift in psychological ownership, in turn, can substantially impact group processes within AR workspaces. In this context, “group processes” refers to the dynamics and interactions within a group working towards a common goal. It encompasses aspects such as gain of shared knowledge, quality of communication, communication means, understanding & expression, ease of referencing & planning, interpersonal awareness, and awareness over time [10]. Dynamics of psychological ownership can significantly impact group processes in AR workspaces. The sense of ownership over virtual spaces in AR can affect group dynamics, decision-making processes, and overall collaboration outcomes. Overall, understanding the interplay between psychological ownership and group processes in AR workspaces represents a burgeoning area of research. Therefore, further research is needed to explore a conceptual framework for the motives and role of psychological ownership in AR shared workspaces.

3 RESEARCH OBJECTIVES AND APPROACH

In my dissertation, I plan to systematically investigate the motives and role of IPO and CPO in remote collaboration around physical objects. First, to investigate how the collaborative functionality affects the sense of IPO and CPO in AR workspaces (RQ1), and second, to explore how IPO and CPO mediate the impact of the collaborative functionality on group processes [10] such as shared knowledge, communication quality, ease of referencing, and interpersonal awareness (RQ2). These research questions are rooted in the complex dynamics of ownership and control uncovered in the prior study and aim to delve deeper into how IPO and CPO are shaped and how they affect collaboration outcomes. To explore

the role of psychological ownership in AR remote collaboration, I will utilize the system I developed called HoloMentor [27] to provide collaborative functionality, specifically designed to control and modification of annotations. I will conduct a controlled within-subject experiment comparing three distinct conditions: 1) both the remote worker and the local worker have the ability to control and modify AR annotations, 2) only the remote worker has the ability, and 3) only the local worker has the ability. I designed a collaborative planning task where collaborators must constantly share and then update their contributions. For example, in collocated planning tasks, moveable objects or erasable whiteboard markers allow for continuous updating to occur. On the other hand, in distributed planning tasks, digital platforms are particularly useful for updating, removing, or undoing virtual annotations. Thus, a simulation task was designed to plan a map focusing on remote collaboration around physical objects and the need for digital annotation modification. In the task, a sense of psychological ownership over the physical or digital objects may play a substantial moderating role in how smoothly the collaborators can coordinate toward a decision. During the collaborative planning task, all interactions will be rigorously documented through audio-visual recordings and screen captures. Researchers will take notes to complement these recordings, offering additional context and observations that may not be readily noticeable in the recordings alone. To delve into the underlying psychological constructs, I will use a series of curated questionnaires: 1) IPO and CPO questionnaire, inspired by Martinaityte’s research [18], 2) five dimensions of psychological ownership questionnaire, adapted from Poretski’s study [23], and 3) group processes questionnaire from Convertino’s research [10]. To conclude the data collection, participants will be presented in semi-structured interviews after tasks. These discussions are intended to delve beyond surface-level interactions, seeking the participants’ deeper reflections and insights that emerged during their engagement with the HoloMentor system.

RQ1: How does the ability to modify annotations impact the sense of IPO and CPO in AR workspaces?

To explore the first research question, the analysis will focus on understanding the nuanced interplay between IPO and CPO within AR workspaces. My analysis will draw on frameworks established by Martinaityte [18] and Poretski [23], which probe into the complexities of ownership perception. The investigation will engage ordinal logistic regression to understand the relationships between collaborative functionality and IPO and CPO. Additionally, moderated regression will be used to analyze the interplay between Poretski’s psychological ownership dimensions across various annotation scenarios. Complementing these quantitative methods, a thematic analysis will be applied to the qualitative data, extracting themes related to ownership discussions and behaviors in the context of annotation modification.

RQ2: How do IPO and CPO mediate the impact of the collaborative functionality on the group processes in AR workspaces?

The second research question will explore the effect of IPO and CPO as mediators on the group processes [10] in AR workspaces.

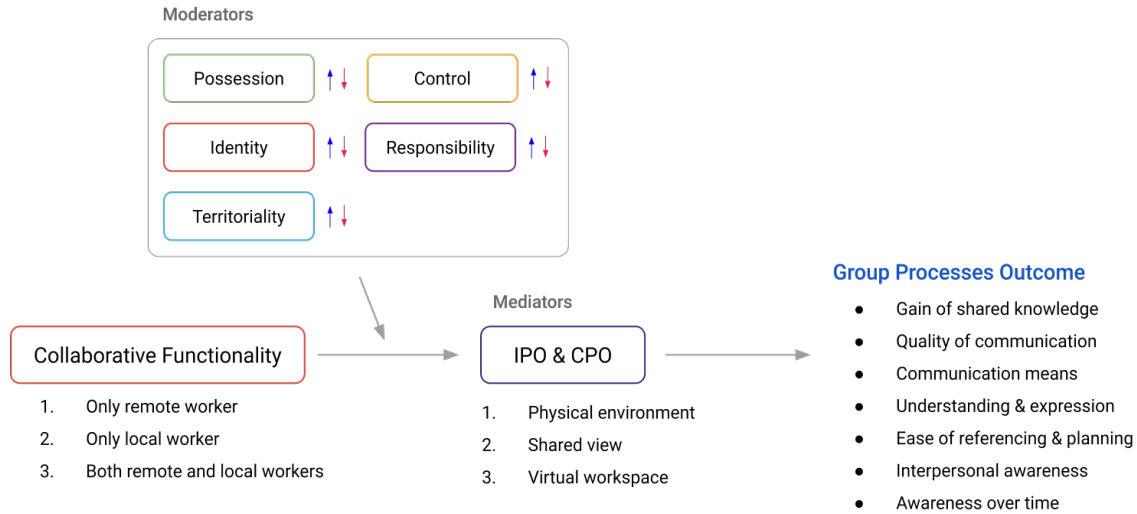


Figure 1: Conceptual Model of Psychological Ownership in shared AR. Collaborative functionality is categorized into three conditions: 1) exclusive control by the remote worker, 2) exclusive control by the local worker, and 3) joint control. Psychological ownership is evaluated across three settings: the physical environment, the shared view, and the virtual workspace. Five dimensions of psychological ownership act as moderators, with group processes measured as outcomes of collaboration.

The quantitative component will use mediation analysis to investigate how IPO and CPO mediate the impact of collaborative functionality on group processes—such as shared knowledge, communication quality, and interpersonal awareness—within the conditions of the HoloMentor system. In parallel, thematic analysis will be applied to qualitative data, identifying and interpreting themes that encapsulate participants’ reflections on psychological ownership and its effects on group processes.

4 RESULTS AND CONTRIBUTIONS

My dissertation research aims to contribute a conceptual framework (Figure 1) elucidating the functionality changes, psychological ownership, and group processes relationships within AR shared workspaces. This investigation contributes both theoretical understanding and practical applications. Firstly, my research sheds light on how different collaborative functionality in AR workspaces affect IPO and CPO. Additionally, my study unveils the complicated relationship between functionality changes and group processes, mediated by IPO and CPO. Secondly, my study provides practical insights that lead to guidelines for best practices in designing mentoring tools within AR workspaces. These functional outcomes offer implications for design, demonstrating how AR systems can be tailored to foster a sense of ownership and collaboration. This aspect of my research serves as a bridge between theoretical findings and real-world applications, providing actionable recommendations for AR developers, UX designers, and researchers. Together, these contributions represent a holistic approach to understanding the relationship between functionality and psychological ownership in AR.

5 EXPECTED NEXT STEPS AND OPEN QUESTIONS

Building on version 1 of the HoloMentor system, I am currently developing the collaborative functionality that allows local and remote workers to create and edit annotations at the same time. I plan to collect data through user testing in early 2024. The goal of participating in the CHI 2024 in-person Doctoral Consortium is to gain feedback on methods and contribution of my work to the greater field of HCI and, more narrowly, social AR/VR. One open question I face is how to effectively combine quantitative surveys with qualitative observations and interviews to fully grasp the concept of psychological ownership in AR collaboration. I am open to learning methods for integrating these results systematically from the feedback of experienced HCI researchers and the fresh perspective of other graduate students.

6 DISSERTATION STATUS AND LONG-TERM GOALS

I am a 5th year Ph.D. student in the Human-Centered Computing program at the University of Maryland, Baltimore County (UMBC), advised by Dr. Helena Mentis. I expect to defend in Summer 2024. To date, my thesis-related work has one paper [27] published at GROUP2023 and one paper under review. Seo et al. [27] proposed the AR HMD-based collaborative system that provides shared workspaces in the physical environment, which we called HoloMentor. The paper under review investigated the nuances of psychological ownership in teleinstructional AR workspaces, focusing on the roles of control, responsibility, and task nature. I would like to complete my thesis with feedback from the Doctoral Consortium. I have not participated in a Doctoral Consortium. At the time of the CHI 2024 DC, I will have just finished collecting data and the

DC will be well-time to provide me with insights on data analysis, framing, contribution, and next steps in my research career. After graduation, I will seek industry research positions in companies that contribute to developing AR systems and improving the AR experience. My long-term goal is to use my skill set in designing, implementing, and evaluating collaborative functionalities to improve the social interaction of AR interfaces and applications.

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